

Exercise on Producer theory

Question 1:

Assume a firm with the following production function:

TP (Q)	L	MP	FC	VC	TC	MC	ATC	AVC
0	0				80			
1	8							
2	12							
3	17							
4	23							
5	30							
6	38							
7	48							

It is known that one unit of labor (L) costs 10 USD.

1. Complete the table.
2. What is the minimal price at which the firm will produce in the short term?
3. What is the minimal price at which the firm will produce in the long term?
4. It is known that the price is 70 USD. What would be the production of the firm?
5. What would be its profit in the short term?
6. What would be its profit in the long term?
7. Will the firm produce in the short term?
8. Will the firm produce in the long term?

Question 2:

There is a market with 8 workers who should be allocated between two fields with the following production function:

Field A

L	TP
0	0
1	5
2	9
3	12
4	14
5	15

Field B

L	TP
0	0
1	3
2	7
3	10
4	12
5	13

- a. How many workers would work on each field if it is known that the price of one produced unit on field A is 2USD and the price of one produced unit on field B is 1USD?
- b. What would be the salary of each worker?
- c. What would be the profit of the owners of each field?